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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 04/02/2018 |
| **PI Name:** | Vanapalli, Siva | | |
| **Project Title:** | Determining Muscle Strength in Space-Flown Caenorhabditis elegans | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Invertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Reproductive Biology<br>(2) Developmental Biology<br>(3) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **Zip Code:** | 79409-0001 | **Congressional District:** | 19 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 Space Biology Flight (ILSRA) NNH14ZTT002N |
| **Start Date:** | 05/19/2015 | **End Date:** | 08/31/2019 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed., 3/5/18)<br>NOTE: Extended to 5/18/2018 per F. Hernandez/ARC and NSSC information (Ed., 6/27/17) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Blawzdziewicz, Jerzy Ph.D. ( Texas Tech University ) | | |
| **Grant/Contract No.:** | NNX15AL16G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | A major impediment to long-duration space travel is loss of muscle mass and strength during spaceflight. Surprisingly, the small worm Caenorhabditis elegans is a good, cost-effective model for studying the effects of spaceflight upon muscle. Our past experiments on Earth and on the International Space Station (ISS) have shown that the expression of key muscle genes that encode members of a muscle attachment complex is reduced in C. elegans during spaceflight. A key open question is if and how these changes in gene expression result in reduced muscle strength in response to spaceflight. While reduced force production is documented for astronauts, it is yet to be demonstrated for C. elegans. To address this critical knowledge gap, we developed a tiny new device that enables continuous measurement of worm muscle forces. Using this device, on Earth, we have shown that defects in muscle attachment complexes lead to decline in worm muscle strength. Motivated by these past flight data and ground results, we propose to (a) demonstrate the flight-readiness of our miniaturized device for quantitative characterization of muscle strength of C. elegans, (b) test our hypothesis, based on past flight data, that muscle strength decreases in flight as a result of altered muscle gene expression in the worm, and (c) quantify muscle strength loss in spaceflight across multiple worm generations. Successful measurements of muscle strength in flight will make C. elegans a much more complete and comprehensive genomic model for muscle physiology, enabling meaningful translation of discoveries made in C. elegans to astronauts. Our proposed studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight. |
| **Task Progress:** | Following our suggested ground testing timeline, we have made significant progress on many aspects of our project, including culturing of selected nematode strains in the axenic media, making strength measurements of wild-type and mutant nematodes cultured in the axenic media, and culturing of nematodes in axenic media in fluorinated ethylene propylene (FEP) bags. We also pursued various spin-off projects related to our original proposal during the lag in hardware acquisition from BioServe.<br>Previously we described successful culturing in the axenic media with the wild-type animal as well as the unc-112 and unc-52 muscular mutants suggested for use in spaceflight. Based on recent experiments, however, we believe the dys-1 mutant strain is also of interest. The dys-1 gene of C. elegans has been studied previously during spaceflight. Expression of dys-1 significantly increases during spaceflight, and a more recent study has shown that gene expression is not affected as much in dys-1 mutants as in wild-type worms during spaceflight. The previous research with dys-1 in space as well as new findings from our own lab led us to also start culturing dys-1 in the axenic media as well.<br><br>Using an optimized NemaFlex platform for measuring the strength of C. elegans, we previously obtained strength measurements for age-synchronized wild-type animals at various time points. During the flight, animals will not be age-synchronized but rather in a mixed population of worms of all different ages. Thus, we need to be able to select for worms of a desired approximate age. The criteria for selecting nematodes of the appropriate developmental stage (young adult) were tested on wild-type and unc-52 animals in these devices. The young adult animals were selected based on the fact that they contain oocytes, making them distinct from larval animals, and also that their diameters have not reached near the maximum limit of growth yet as seen in older adult animals. To test the robustness of our selection criteria, we compared young adult animals selected from the mixed population inside the device with animals of the predicted same age from an age-synchronized population of young adults. The strength values for wild-type age-synchronized animals and mixed population animals agreed well, with no significant difference between the two, indicating that the selection criteria work well for wild-type animals. However, there was some disparity between the unc-52 animals that were age-synchronized and mixed, and the mixed-age unc-52 mutants were not discernibly different from wild-type animals. This indicates that unc-52 strength may be difficult to discern from wild-type animals since the mutation does not severely affect the muscle function. Unc-52 animals of the age-synchronized population are also skinnier on average than the mixed population, meaning that the animals we selected to analyze in the mixed population may have been slightly too old; this can be fixed by setting a diameter cutoff to exclude animals that are too fat. We expect that unc-112 or dys-1 will be better mutants to study in this context because they are much weaker than both wild-type and unc-52 in our E. coli-based studies.<br><br>Because we had not received BioCells from BioServe during 2017, we decided to explore alternate culturing vessels. We utilized the Origen PermaLife Cell Culture Bags-PL07 (12mL maximum volume). We injected 2500 larval stage 1 animals into the culture bag, and within 7 days (the approximate time to reach adulthood), the population size had grown to 15,000, which was comparable to animals grown in a petri dish. This indicates that there was proper oxygen exchange and favorable conditions for animals to grow. Culture bags have been utilized in the past for spaceflight studies with C. elegans, so this type of FEP bag is a good option for a culture vessel during spaceflight. We expect that animals will also culture well in the VueLife bags that we recently acquired from BioServe.<br><br>During the downtime in which we were not sent the hardware we needed to continue our ground studies for the spaceflight experiment, we pursued three main side projects: 1) investigating dystrophin mutants for strength defects and screening drugs to improve strength, 2) looking at effects on muscle strength from starvation or diet restriction (axenic media) when transferring worms away from E. coli at the beginning of adulthood, and 3) analyzing fatty acid composition using FTIR (Fourier transform infrared spectroscopy) in worms fed axenically.<br><br>The dystrophin protein is important for proper muscle function in both humans and C. elegans. In humans, mutations in the dystrophin gene can lead to severe muscle disorders including Duchenne Muscular Dystrophy. Additionally, the dys-1 gene in C. elegans plays an important role in muscle function and is differentially regulated during spaceflight. We hypothesized that dys-1 mutants of C. elegans would be detectably weaker in our NemaFlex devices, which we did observe in the BZ33 strain of dys-1 for adults older than 1 day. We also tested whether drugs known to improve certain aspects of muscle health could alleviate the strength loss in dys-1 mutants. Both prednisone, the standard compound for treating muscular dystrophy symptoms in humans, and melatonin improved muscle strength in C. elegans dys-1 mutants. This indicates dys-1 as an interesting mutant for studying loss of muscle strength related to diseases and spaceflight, and it also validates NemaFlex as a useful platform for testing the efficacy of compounds in treating the loss of muscle strength. This platform may be useful in future studies for testing drugs that could help to prevent loss of muscle strength in astronauts.<br><br>To better understand how muscle structure and function changes with different diets, we investigated how animal |

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| | strength changes when transferred from an E. coli diet at the end of development (Day 1 Adult) to either a starvation condition of buffer or to axenic media. Animals kept on the E. coli diet through Day 2 of adulthood continued to grow in strength, while those transferred to buffer or axenic media lost strength. This further validates that axenic media is distinct from a starvation condition and may better resemble a dietary restriction condition. This also demonstrates that muscle strength is strongly affected by the animal's diet.<br><br>Finally, we explored using Fourier Transform Infrared (FTIR) imaging to detect changes in C. elegans' diet. We utilized the axenic media with and without a supplemented fatty acid called Eicosapentaenoic acid (EPA). The use of axenic media helped to prevent confounding the data by the fatty acids present in E. coli. We showed that distinct peaks representing fatty acids were present in wild-type animals but absent or reduced in fat-3 and tub-1 mutants that are not able to synthesize certain fatty acids. Supplementing the diets of fat-3 and tub-1 animals with EPA changes the FTIR spectra to indicate changes in chemical composition of the animal. |
| **Bibliography Type:** | Description: (Last Updated: 10/20/2025) |
| **Articles in Peer-reviewed Journals** | Bouyanfif A, Liyanage S, Hewitt JE, Vanapalli SA, Moustaid-Moussa N, Hequet E, Abidi N. "FTIR imaging detects diet and genotype-dependent chemical composition changes in wild type and mutant C. elegans strains." Analyst. 2017 Dec 4;142(24):4727-36. https://doi.org/10.1039/c7an01432e ; PubMed PMID: 29139484 , Dec-2017 |
| **Articles in Peer-reviewed Journals** | Bilbao A, Patel AK, Rahman M, Vanapalli SA, Blawzdziewicz J. "Roll maneuvers are essential for active reorientation of Caenorhabditis elegans in 3D media." Proc Natl Acad Sci U S A. 2018 Apr 17;115(16):E3616-E3625. Epub 2018 Apr 4. https://doi.org/10.1073/pnas.1706754115 ; PubMed PMID: 29618610; PubMed Central PMCID: PMC5910807 , Apr-2018 |